Sudhir Kumar Singh

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Summary

Technical leader and software generalist with around 12 years of experience solving complex problems across diverse technologies and domains, bringing a problem-first mindset to adapt and solve, and a passion for AI to shape the future.

Education

Rochester Institute of Technology — Rochester, NY
Master of Science, Computer Science — Aug 2018 - May 2021

Oriental College of Technology — Bhopal, MP
Bachelor of Engineering, Information Technology — Jul 2006 - May 2010

Skills (worked with Tech)

GenAI	: LangChain, LangGraph, MCP, Embeddings, pgvector, LLMs, RAG, Agents, Evals, Guardrails
AI core	: PyTorch, scikit-learn, ML, NLP, llama.cpp, Claude Code
Coding	: Python, FastAPI, Flask, Java, Spring Boot, Spring WebFlux, React, Bash
Database/Tools	: SQL, NoSQL, VectorDB, Postgres, Cassandra, Kafka, Git, GitHub, Gradle, Maven, Jenkins
Infra	: AWS, Docker, Kubernetes, ArgoCD, ELK, Prometheus, Grafana, CI/CD, DevOps
Others	: Distributed Systems, Microservices, Databases, REST, OpenAPI, Jira, HIPAA
Leadership	: Technical Leadership, Mentoring & Coaching, Cross-functional Collaboration, Agile Delivery

Projects

Interview Prep Hub *(Java, Spring boot, Python, FastAPI, Postgres, pgvector, llama.cpp, Next.js)* Jun 2026 - Present

- Building a unified learning platform covering system & AI/ML system design interviews with 50+ real-world HLD case studies, hands-on problem-solving and ML coding practice, and a RAG-powered AI Tutor for grounded Q&A.
- Designing a microservices backend (sandboxed code-execution workers, kafka-based queues) with an LLM-as-judge pipeline for automated test-case validation, code review, scoring, and personalized learning recommendations.

TheAIAtlas *(Python, FastAPI, Postgres, pgvector, LiteLLM, llama.cpp, Next.js)* Jul 2026 - Present

- Building a one-stop AI intelligence platform: models with architecture-level fingerprints, benchmarks, research papers, news, hardware, and an AI-authored weekly newsletter, replacing the scattered sources engineers juggle today.
- Architecting an agentic pipeline (crawl, extract, LLM synthesis, semantic indexing) autonomously refreshing the knowledge vault and pgvector database powered by a self-hosted open-source LLM.

miniAgent *(Python, FastAPI, Asyncio, LangChain, LangGraph, Google ADK, LiteLLM)* Jul 2026

- Built a lightweight, provider-agnostic orchestrator mini SDK for agentic workflows that swaps model providers across AI agents without any code changes, eliminating vendor lock-in.
- Supports both linear pipelines and graph-based workflows with conditional routing and asynchronous tool execution for multi-agent workflows and reasoning.

Classifying similar language varieties *(Python, scikit-learn, NLP, NLTK, NumPy)* May 2019

- Developed machine learning models to classify similar languages and language varieties using the DSL dataset, leveraging TF-IDF character and word n-gram features with Naive Bayes and SVM models, achieving 88% accuracy.

Work Experience

Walmart Inc — Sunnyvale, CA
Senior Software Engineer (via Compunnel) — Nov 2023 - Oct 2025

- Built and operated core services of Walmart's item-eligibility platform, 40+ high-throughput microservices deriving eligibility and restrictions for offers and products at store and fulfillment-center levels, powering every shopping journey on walmart.com across US and global markets. **(Java17, WebFlux, Cassandra, Kafka, K8s, Cursor)**
- Designed and scaled 2 microservices from scratch to process 10K+ events/min and evolved the platform's inbound and outbound services handling 300M+ events daily on multitenant Cassandra. **(Spring WebFlux, Kafka)**
- Architected backend enhancements with engineering, data science, and product teams to unify 1P/3P shipping workflows, improving delivery promise accuracy and cutting late deliveries by ∼15%.
- Led architectural and design discussions, sprint ceremonies, code reviews, and engineering standards across the team, and mentored 4-5 junior engineers.
- Drove reliability improvements across monitoring, availability, and operational support, serving in on-call rotations and war rooms for production incidents.

Strateos Inc Menlo Park, CA
Software Engineer III Jul 2021 - Jun 2023

- Designed and deployed Asset-management service components and UI (cloud lab) to remove tedious manual orchestration and management (Ansible scripts) of labs, workcells, and devices. **(Java17, Spring Boot, React)**
- Developed components using an event-driven architecture to compile a scientific intent into device commands; coordinating scheduling, execution, and reporting to reduce scientist time by ∼90%. **(Scala, Akka, RabbitMQ)**
- Led a 2-developer effort and drove the implementation of an automated colony-picking system using a robotic arm and computer vision, achieving ∼10× higher throughput than manual inspection. **(Python, OpenCV)**
- Owned end-to-end delivery of multiple product feature epics, from design through production deployment, collaborating with scientists, product, and platform teams.
- Conducted technical interviews and contributed to engineering hiring decisions to support team growth.

Center On Access Technology, RIT Rochester, NY
ML Software Engineer - (Part-time) Jan 2021 - May 2021

- Built language and and error models using n-gram techniques on ASR (speech-to-text) data to improve hybrid captioning accuracy for a hard-of-hearing classroom. **(Python, NLTK)**
- Developed the frontend and backend services to deploy, store, and serve ML models on AWS, enabling interactive evaluation and performance analysis of captioning models. **(Python, Flask, React, AWS)**
- Experimented and evaluated many NLP models, including transformer-based architectures in Hugging Face, to improve captioning accuracy on ASR datasets. **(PyTorch, BERT)**

Philips India Limited Bangalore, India
Senior Software Engineer Feb 2015 - Jul 2018

- Developed reliable REST APIs for the HealthSuite Digital Platform (HSDP) to enable secure exchange of patient health information between internal and external systems using HL7 and FHIR standards.
- Interfaced with multiple services like the central data repository, oncology service, and external hospital systems services securely with token-based authentication (OAuth) using the internal IAM service.
- Served as technical lead for 8 developers, facilitated sprint planning and scrum ceremonies, conducted design reviews, resolved cross-team dependencies, and drove the Oncology integration project across two major platform releases.

CliniSys Solutions Limited Bangalore, India
Software Engineer Jun 2011 - Feb 2015

- Owned a product's end-to-end lifecycles with a team of 4 developers, covering requirements, design, development, deployment, and client interactions, while delivering 3 parallel product releases with **zero** client-reported issues.
- Worked directly with clients on custom feature requirements, implementation, deployment, and validation. Also, supported on-call rotation, collaborated with L1 support and clients to resolve **business-critical escalations**.
- Developed product features across laboratory and pathology desktop and web solutions, including patient registration, results management, reporting, and HL7 message exchange for inpatient and outpatient workflows.

Awards

Best Innovative App (Android) Philips India Limited
Won second-best innovative app award for developing a DigitalOPD app as part of a Philips hackathon. Feb 2018

Circle of Excellence'14 CliniSys Solutions Limited
Recognized as the top 5 outstanding performers of the company across the global region. Jan 2014